

Rajat Savdekar

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EDUCATION

University of Southern California, Los Angeles Master of Science, Computer Science Coursework: Algorithms, Database Systems, Machine Learning, Web Technology, VR / XR Development	Aug 2024 – May 2026 GPA: 3.63
University of Mumbai, Mumbai Bachelor of Engineering, Computer Engineering Coursework: Operating Systems, Computer Networks, Distributed Systems, Natural Language Processing	Aug 2019 – May 2023 GPA: 3.7

SKILLS

Languages & Core: Python, TypeScript, JavaScript, SQL, Java, C++, HTML, CSS
Backend & Data Systems: FastAPI, Flask, Node.js, REST APIs, WebSockets, PostgreSQL, Redis, MongoDB, LevelDB, Celery
AI Infrastructure & Cloud: PyTorch, Scikit-learn, OpenAI API, AWS, GCP, Azure, Docker, Terraform, CI/CD, React, Next.js

PROFESSIONAL EXPERIENCE

Software Developer - Systems & AI Easley-Dunn Productions, Inc. - Mediverse (VR Surgical Simulation)	Aug 2026 – Present Torrance, CA
- Reduced AI narration inference cost by 60% and replay turnaround from 22 to 4 minutes by implementing a warm 4-bit quantized 7B worker, batched speech synthesis, and content-hash caching. - Built a human-in-the-loop grounding gate for AI surgical narration, eliminating ungrounded operative notes by validating each generated sentence against the source event log. - Raised replay upload success from 82% to 99.6% across 150+ sessions by replacing 100 MB in-memory SQLite blobs with streamed object storage and Postgres metadata.	
Software Engineer - Backend & Distributed Systems JFSS – Agritech Supply Chain Platform	May 2023 – May 2024 Jalgaon, India
- Decomposed a legacy PHP monolith into Python Flask microservices backed by PostgreSQL with cron-based state sync, cutting query latency by 40% across 1,000+ daily truck transactions. - Doubled peak harvest write throughput by architecting an isolated write path that eliminated 10–15 minute MyISAM lock freezes, sustaining a season where platform volume grew from \$100K to \$200K. - Delivered a web procurement platform with REST APIs and live analytics dashboards, migrating 150+ daily clerks across 5 warehouses off manual logs.	
Software Engineer Intern - AI & Data Systems Softaid Computers	Jan 2023 – Apr 2023 Jalgaon, India
- Evaluated domain fine-tuning against semantic RAG architectures, adopting vector search with document chunking to eliminate ungrounded responses and lower query processing cost by 35%. - Reduced frontend telemetry latency by 80% by migrating from polling endpoints to a real-time React and Tailwind interface backed by persistent WebSocket connections.	

SELECTED PROJECTS

MockPad – Real-Time Collaborative Code Editor	[Live] [GitHub]
- Architected a real-time collaborative editor on Yjs CRDTs over custom WebSockets, sustaining sub-100ms sync latency for 30 active concurrent users on a single \$6/month droplet, and have around 50 real user. - Cut idle server memory by 80% while ensuring seamless state restoration whenever a collaborator rejoins.	
Caliberate – LLM Code-Eval Harness	[GitHub]
- Built a sandboxed eval harness running patches across isolated Docker containers, preventing task loss using a PostgreSQL SKIP LOCKED queue with lease-based crash recovery. - Automated regression testing in CI pipelines to automatically reject low-quality model code changes.	
CloudBridge – AI Cloud Architecture Advisor & Topology Engine	[GitHub]
- Engineered a FastAPI synthesis engine turning natural language into verified AWS topologies, blocking designs that violate Well-Architected rules or IAM least-privilege before export. - Delivered interactive infrastructure visualization that provides real-time monthly spending estimates.	

PUBLICATIONS & HACKATHONS

Research Publications. Four undergrad papers; GLCM cited 10x across Nature, Springer, IEEE.	[GitHub]
Stitch – Autonomous CI Repair Agent. Aug 2026. OpenAI Build Week, Developer Tools track.	[GitHub]
Aura – visionOS Accessibility Assistant. Aug 2025. 2nd place of 24-hour hackathon.	[Demo]
TrojanMind – Campus Copilot. Anthropic Hackathon. AI academic copilot for USC students.	[Demo]
GSAP Labs – Iron Man & F1. Scroll-driven web sequences using Astro, GSAP, and Webflow.	[Demo]